rag -- 2026-11-14

5 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

Worth reading in full

[P1] DISCERN: Preserve at Write, Resolve at Read for Persistent LLM-Agent Memory -- DISCERN directly tackles retrieval quality, entity resolution, and calibrated abstention--key aspects of grounding and faithfulness in RAG--making it the most informative single paper for the stated topic.

Themes

Several papers address core challenges in retrieval-augmented generation (RAG) from different angles. DISCERN proposes a bi-temporal provenance graph with a read-time tool-driven navigation loop that dramatically improves answer correctness (macro 0.861) and calibrated abstention (F1 0.924) over eight baselines, highlighting the importance of explicit entity resolution and temporal grounding in persistent LLM agents [P1]. In contrast, Edge-Native Multimodal RAG focuses on system-level constraints, showing that hardware-aware NF4 quantization combined with structured few-shot prompting can fit a multimodal RAG pipeline into 8.37 GB VRAM while restoring mathematical reasoning accuracy from 35 % to 92 % on 200 trials, underscoring trade-offs between latency, stability and model size on consumer GPUs [P2]. Both works share a design-centric view of RAG but differ: DISCERN optimizes memory semantics and decision-making, whereas Edge-Native optimizes hardware efficiency and multimodal integration.

Evaluation of retrieval components also appears across the set. CaST introduces a calibrated early-termination rule for constrained vector search that stays within about 13 % of the class-optimal cost while guaranteeing a user-specified error risk, and competing calibrated rules spend between 6.2 % and 70.6 % more visits than CaST on million-item datasets [P3]. This contributes a principled cost-accuracy trade-off for the retrieval back-end that complements DISCERN's focus on answer-level correctness.

AutoOntic automatically induces a hierarchical ontology from raw text using LLM-driven typed triplet extraction and Contriever clustering, achieving 85.87 % information-retention accuracy, which rises to 88.00 % when initialized with a Wikidata-seeded hierarchy, and improves downstream retrieval to 88 % on the MINE-1 benchmark [P4]. BharatSynthQA provides a new culturally-specific passage-grounded QA benchmark that shows LLM performance improves markedly when relevant passages are supplied, for example Factoid F1 increases from roughly 0.2 to roughly 0.75, highlighting grounding challenges in current models [P5].

Overall, the collection shows a convergence on three themes: (1) richer memory and grounding mechanisms (DISCERN, AutoOntic), (2) system-level efficiency and hardware-aware design (Edge-Native), and (3) rigorous retrieval evaluation with calibrated guarantees (CaST). The new dataset (BharatSynthQA) exposes real-world grounding failures, motivating further integration of these advances.

5 theme sentence(s) checked against the paper summaries; 3 corrected, 0 removed.

At a glance

[P1] DISCERN: Preserve at Write, Resolve at Read for Persistent LLM-Agent Memory · relevance 0.37
DISCERN introduces a "preserve-at-write, resolve-at-read" memory system that outperforms eight baselines on both answer correctness and calibrated abstention in synthetic enterprise and out-of-domain benchmarks.

[P2] Edge-Native Multimodal RAG: Hardware-Aware Quantization and Architectural Optimization for Consumer GPUs · relevance 0.28
The paper demonstrates a hardware-aware quantization and modular multimodal RAG design that fits

within an 8.37 GB VRAM budget on consumer-grade Turing GPUs, using NF4 quantization and prompt engineering to recover reasoning accuracy and cut latency.

[P3] Calibrated Early Termination with Bounded Optimality Gaps for Constrained Vector Search · relevance 0.27
CaST is a calibrated early-termination rule for constrained vector-search that provably stays within about 13 % of the best possible cost in its monotone class while guaranteeing a user-specified error risk.

[P4] AutoOntic: Automatic Ontology Induction from Text for Knowledge Graph Construction · relevance 0.26
AutoOntic automatically builds a hierarchical ontology from raw text and uses it to improve knowledge-graph extraction and downstream retrieval.

[P5] BharatSynthQA: A Context-Grounded Question Answering Dataset on Indian Temples and Religious Heritage · relevance 0.21
BharatSynthQA is a newly released, passage-grounded QA dataset on Indian temples that reveals current LLMs perform poorly on this cultural domain without source passages, but improve markedly when passages are provided.

[P1] DISCERN: Preserve at Write, Resolve at Read for Persistent LLM-Agent Memory

Chirag Ahuja, Vladislav Tanov, Ramcharan Kakarla, Jonathan Taws · arXiv 2611.15729v1 · https://arxiv.org/abs/2611.15729v1 · relevance 0.37

TL;DR: DISCERN introduces a "preserve-at-write, resolve-at-read" memory system that outperforms eight baselines on both answer correctness and calibrated abstention in synthetic enterprise and out-of-domain benchmarks.

Problem
Enterprise LLM agents accumulate contradictory and time-varying facts across many sessions, leading to entity confusion and stale answers; existing work treats this as a pure retrieval problem, which leaves resolution (identifying the correct entity, current time-state, and deciding when to abstain) unsolved [1][2].

Method
DISCERN builds an immutable per-actor event log and a strictly additive bi-temporal provenance graph that never merges confusable entities or deletes superseded facts [2][3]. At read time, a zero-LLM tool-driven agentic loop navigates this graph hierarchically (profile -> entity/session -> fact -> transcript) and commits one of four gated terminal actions (ANSWER, UPDATE, CLARIFY, ABSTAIN) [2][3]. Evaluation uses a four-judge council to score answer correctness and the answer-versus-abstain decision [2][5].

Key results
- On the ENT-310 benchmark (210 answerable, 100 unanswerable queries), DISCERN achieves a macro answer-correctness of 0.861, surpassing the strongest baseline (Extract-Retrieve 0.806) [5].
- Entity-resolution score is 0.800, whereas no baseline exceeds 0.40 [5].
- Abstention performance: F1 0.924 and false-decline 0.029, the best among eight baselines (next best HippoRAG2 F1 0.802) [5][2].
- Pillar scores on ENT-310: retention 0.929, currency 0.900, contamination 0.817, entity-resolution 0.800 [5].
- Out-of-domain LoCoMo transfer (1,986 questions) yields macro answer-correctness 0.663 (vs. HippoRAG2 0.638) and abstention F1 0.820 with false-decline near 0.02-0.03 [5].
- Ablation shows the read-time navigation policy raises accuracy from a 0.33 "tools-off" floor to 0.89 [2].

Limitations
The paper's excerpts do not detail limitations beyond noting synthetic benchmark construction and that most baselines are single-seed while DISCERN uses three-seed means; no real-world deployment results

or analysis of scalability are provided in the cited text. [6]

Sources

[1] arXiv 2611.15729v1.pdf, arXiv abstract

[2] arXiv 2611.15729v1.pdf, page 2

[3] arXiv 2611.15729v1.pdf, page 4

[5] arXiv 2611.15729v1.pdf, page 7

[6] arXiv 2611.15729v1.pdf, page 9

[P2] Edge-Native Multimodal RAG: Hardware-Aware Quantization and Architectural Optimization for Consumer GPUs

Zahraa Selim, Menna Hamed, Wesam Ahmed, Sohyla Said et al. · arXiv 2611.14924v1 · https://arxiv.org/abs/2611.14924v1 · relevance 0.28

TL;DR: The paper demonstrates a hardware-aware quantization and modular multimodal RAG design that fits within an 8.37 GB VRAM budget on consumer-grade Turing GPUs, using NF4 quantization and prompt engineering to recover reasoning accuracy and cut latency.

Problem
Deploying retrieval-augmented generation on edge devices is hampered by memory and bandwidth limits, as a 7 B parameter model in FP16 consumes ~14 GB VRAM, leaving insufficient headroom for vector indices and KV caches on typical 12-16 GB GPUs [1][2]. Quantization techniques that improve throughput can cause severe context instability on Turing architectures, especially for sequences longer than 2048 tokens [1][2][5].

Method
The authors benchmark three post-training quantization schemes (GPTQ, NF4, AWQ) on an NVIDIA Tesla T4 to assess latency, throughput, and stability [3][4][5]. They select NormalFloat-4 (NF4) for its robust long-context performance and integrate a decoupled Florence-2 visual module, adding only ~430 MB VRAM [1][3]. Structured few-shot prompting is employed to mitigate quantization-induced reasoning loss, and prompt-induced search-space pruning is used to reduce generation latency [1][3][6][7].

Key results
- GPTQ achieves the highest theoretical throughput (51.65 tokens/s) and fastest time-to-first-token (43 ms) but exhibits context collapse beyond 2048 tokens [5].
- NF4 runs at 81.37 ms/token (12.29 tokens/s) with a TTFT of 178 ms, providing stable inference [5].
- The full multimodal pipeline occupies 8.37 GB VRAM, well within the 12-16 GB envelope of consumer GPUs [1][5][6][7].
- Visual reasoning adds only 430 MB VRAM overhead [1][3].
- Structured few-shot prompting restores mathematical accuracy from 35 % to 92 % across 200 trials [1][3][6][7].
- Prompt-induced search-space pruning cuts generation latency by 47 % without changing computational intensity (1.88 GFLOPs/token) [1][3].

Limitations
GPTQ's instability limits its use for long contexts on Turing GPUs, and NF4, while stable, is slower than GPTQ [5]. The evaluation is confined to a single Turing GPU (Tesla T4) and a 7 B model; performance on other hardware or larger models is not reported [2][5]. Reasoning restoration is demonstrated only on 200 mathematical trials, with no broader accuracy metrics provided [1][3][6][7].

Sources

[1] arXiv 2611.14924v1.pdf, arXiv abstract

[2] arXiv 2611.14924v1.pdf, page 1

[3] arXiv 2611.14924v1.pdf, page 2

[4] arXiv 2611.14924v1.pdf, page 2

[5] arXiv 2611.14924v1.pdf, page 3

[6] arXiv 2611.14924v1.pdf, page 4

[7] arXiv 2611.14924v1.pdf, page 4

[P3] Calibrated Early Termination with Bounded Optimality Gaps for Constrained Vector Search

Xinyu Zhang, Jingze Li, Baihua Zheng, Weiwei Sun · arXiv 2611.16148v1 · https://arxiv.org/abs/2611.16148v1 · relevance 0.27

TL;DR: CaST is a calibrated early-termination rule for constrained vector-search that provably stays within about 13 % of the best possible cost in its monotone class while guaranteeing a user-specified error risk.

Problem
Vector search over learned embeddings often imposes per-category caps, and graph-based indexes must decide when to stop traversing to balance cost against exactness. Existing learned early-termination methods optimize average recall but do not guarantee that the returned answer is exact with high probability. The paper therefore seeks the cheapest stopping rule whose answer is exact with probability at least $1-\varepsilon$ for constrained k-NN queries [1].

Method
The authors declare a fixed set of observations that any rule may use and restrict attention to monotone stopping rules (rules that stop earlier when the running minimum score drops or the answer becomes full) [3]. Over this monotone class, risk-constrained stopping becomes a linear program on an order polytope; its Lagrangian minimizers form a nested chain of staircase rules that can be computed exactly by dynamic programming [1][2]. The chain is calibrated into the final rule, CaST, using fixed-sequence exact binomial tests that bound the failure probability by ε with confidence $1-\delta$ [1][4]. Simultaneously, Lagrangian weak duality yields a data-dependent lower bound on the class-optimal cost, providing an end-to-end optimality-gap bound within the declared class [2][4].

Key results
- On two datasets of one million items each, CaST's cost is at most 1.132 times the class optimum for risk levels from 0.03 to 0.10, with 95 % joint confidence [1][7].
- At a common risk target $\varepsilon=0.05$, eight other calibrated rules spend 6.2 % to 70.6 % more visits than CaST, and CaST lies within 1.087 × of the lower bound with 87.5 % confidence [2].
- Compared to all other calibrated rules timed, CaST is faster, while the alternatives require 4.8 % to 45.5 % more mean time [1][7].
- A clairvoyant rule that knows each query's completion time would be 2.8 × to 3.3 × cheaper than the empirical class optimum, indicating potential gains outside the monotone class [2].

Limitations
- The optimality-gap bound and risk guarantees apply only to rules restricted to the declared interface; richer policies may achieve larger savings but are not covered [2].
- Guarantees hold for queries drawn i.i.d. from the same distribution as the training samples; a query-distribution shift or index update would require recalibration [7].
The experiments use HNSW graphs [6].
- Larger cost reductions would require rules outside the monotone class, which CaST does not explore [1].

Sources

[1] arXiv 2611.16148v1.pdf, arXiv abstract

[2] arXiv 2611.16148v1.pdf, page 2

[3] arXiv 2611.16148v1.pdf, page 4

[4] arXiv 2611.16148v1.pdf, page 6

[6] arXiv 2611.16148v1.pdf, page 8

[7] arXiv 2611.16148v1.pdf, page 12

[P4] AutoOntic: Automatic Ontology Induction from Text for Knowledge Graph Construction

Alla Chepurova, Aydar Bulatov, Mikhail Burtsev, Yuri Kuratov · arXiv 2611.15467v1 · https://arxiv.org/abs/2611.15467v1 · relevance 0.26

TL;DR: AutoOntic automatically builds a hierarchical ontology from raw text and uses it to improve knowledge-graph extraction and downstream retrieval.

Problem
Retrieval-augmented generation relies on high-quality knowledge graphs, but schema-free LLM extraction pipelines typically produce flat graphs without type hierarchies or domain/range constraints. [1]

Method
AutoOntic first extracts subject-relation-object triplets with free-text entity types using an LLM (typed triplet extraction) [2]. It then canonicalizes relations and entity types by clustering Contriever embeddings with HDBSCAN and adjudicating each candidate cluster with a single LLM call [2][3]. Entity-type canonicalization incorporates relation-argument profiles and a hyponymy test to merge only non-hyponym pairs [2]. A single-parent hierarchy is induced over the canonical types, and domain/range constraints are derived by computing lowest common ancestors in the hierarchy and scoring signatures by support [3][4]. The pipeline uses Contriever sentence embeddings and LLM calls for clustering and verification [2][4].

Key results
On Text2KGBench, over 90% of induced types are semantically aligned with reference classes, and most mismatches reflect differences in granularity rather than unrelated types [5]. Exact-match rates on Wikidata-TekGen range from 53.6 % to 58.6 %; unrelated types stay under 10 % for domains and under 8 % for ranges [4]. On the MINE-1 benchmark, AutoOntic achieves 85.87 % information-retention accuracy, increasing to 88.00 % when initialized with a Wikidata-seeded hierarchy, outperforming KGGen and Wikontic [5]. Retrieval accuracy on MINE reaches 88 % when using the ontology-organized KG [1].

Limitations
The evaluation measures agreement with a reference ontology and downstream retrieval usefulness, not direct extraction precision, and excludes facts not covered by the curated benchmark [6]. It is limited to English-language corpora and existing benchmark ontologies; generalization to other languages remains untested [6].

Sources

[1] arXiv 2611.15467v1.pdf, arXiv abstract

[2] arXiv 2611.15467v1.pdf, page 3

[3] arXiv 2611.15467v1.pdf, page 5

[4] arXiv 2611.15467v1.pdf, page 6

[5] arXiv 2611.15467v1.pdf, page 8

[6] arXiv 2611.15467v1.pdf, page 9

[P5] BharatSynthQA: A Context-Grounded Question Answering Dataset on Indian Temples and Religious Heritage

Riya Naik, Shivam Sharma, Swati Agarwal · arXiv 2611.11952v1 · https://arxiv.org/abs/2611.11952v1 · relevance 0.21

TL;DR: BharatSynthQA is a newly released, passage-grounded QA dataset on Indian temples that reveals current LLMs perform poorly on this cultural domain without source passages, but improve markedly when passages are provided.

Problem
Existing QA resources focus on popular domains and lack coverage of Indian temples and related religious heritage, making it hard to assess or develop culturally aware AI applications in education, tourism, and heritage preservation[1][2].

Method
The authors built a pipeline that (1) seeds a topic-based crawler with authentic temple sources, filters Wikipedia articles by topic similarity (? = 0.2) and Indian location, yielding 11,064 documents after geographic filtering and 7,617 (68.8%) deemed relevant; (2) extracts topics via TextRazor and uses Llama 4 Scout to generate QA pairs in three types (Boolean, Factoid, Explanatory) from 7,617 documents, producing 131,494 pairs; (3) employs DeepSeek V4 Flash 0731 as an LLM judge to filter the pairs, retaining 98,677 high-quality pairs; and (4) adds approximately 300 human-annotated pairs [7][7][3][4][5][7][4].

Key results
- Human evaluation of a 200-pair sample showed 89.83% of questions were formed correctly and about 90% were clear, with 84.16% of answers judged fully or partially correct[1][5].
- Model benchmarking on 1,848 test questions (616 of each type) revealed that without passage context, Factoid token F1 ranged from 0.177 to 0.297, Boolean balanced accuracy from 0.62 to 0.81, and Other ROUGE-L from 0.166 to 0.216; with passage context these scores rose to 0.725-0.766 (Factoid F1), 0.93-0.98 (Boolean accuracy), and 0.459-0.596 (Other ROUGE-L)[6].
- Explanatory questions were the hardest: only 51.3% survived LLM-judge filtering and they had the highest proportion of low-confidence generations (4.04%)[7].

Limitations
The dataset is heavily temple-centred, dominated by Hindu content, and limited to English Wikipedia, leaving other traditions, regions, and Indian languages under-represented[8][8]. All QA pairs are LLM-generated and filtered by a single LLM judge, without extensive human audit or multiple-judge consensus, which may affect quality guarantees[8]. The collection is Factoid-heavy, with fewer Boolean and explanatory items [8].

Sources

[1] arXiv 2611.11952v1.pdf, arXiv abstract

[2] arXiv 2611.11952v1.pdf, page 2

[3] arXiv 2611.11952v1.pdf, page 3

[4] arXiv 2611.11952v1.pdf, page 5

[5] arXiv 2611.11952v1.pdf, page 7

[6] arXiv 2611.11952v1.pdf, page 9

[7] arXiv 2611.11952v1.pdf, page 10

[8] arXiv 2611.11952v1.pdf, page 12